

“GSoC 2026 Project Proposal”

Project: PSLab Add VCD Data Export and Import Functionality

Organization: FOSSASIA

Personal Information:

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1- About Me & My Motivation;

Hi, I'm Mariam, a Business & Commerce student currently transitioning into Data Analysis. I recently started learning Python and am actively building my fundamentals. Coming from a business background, I am particularly interested in data accuracy and consistency. I also want to work on a project where I can handle real-world data instead of just theory; PSLab feels like a great place to start while learning how to contribute to open source. While I have been focusing on Python for data analysis, I am highly motivated to apply these logic concepts using Dart and Flutter to ensure the VCD functionality is integrated directly into the PSLab Android app

2- Problem Statement:

PSLab's Logic Analyzer captures digital signals, but this data is currently limited to the PSLab environment.

Goal: Enable users to export captured data into the VCD (Value Change Dump) format so it can be opened in common tools like GTKWave. This will make it easier to analyze signals outside PSLab and improve usability for debugging and learning. The presence of VCD leaves PSLab users 'isolated' from industry tools such as GTKWave.

3- Proposed Solution & Technical Approach (Dart, Flutter, Python) :

I plan to implement a Python module that supports both exporting to and importing from VCD files. Since I am still strengthening my Python skills, I will focus on writing clean, readable, and well-documented code. I will use Python's unittest library to ensure the exported VCD files are valid.

A. Export Functionality (PSLab to VCD):

```
def save_to_vcd(file_name, timestamps, logic_levels):  
  
    with open(file_name, 'w') as f:  
  
        f.write("$date Today $end\n")  
  
        f.write("$timescale 1us $end\n")  
  
        # This is a simplified single-signal implementation.  
  
        # Future versions will support multiple signals and unique identifiers.  
  
        for i in range(len(timestamps)):  
  
            time = timestamps[i]  
  
            value = logic_levels[i]  
  
            f.write(f"#{time}\n")  
  
            f.write(f"{value}!\n")
```

This initial version focuses on converting timestamps and logic levels into VCD format clearly. I will later expand it to support multiple channels and ensure compatibility with tools like GTKWave.

B. Import Functionality (VCD to PSLab):

For the import feature, I will build a parser that reads VCD files line by line:

- Efficiency:** I will avoid loading the entire file into memory to keep it efficient.
- Reconstruction:** I will reconstruct timestamps and signal values into arrays compatible with PSLab.
- Validation:** I plan to add validation checks to ensure the imported data matches the original structure.

C. Dart's File IO operations to VCD (Dart Syntax):

// Logic to write VCD format in Dart

```
Future<void> saveToVCD(String fileName, List<int> timestamps, List<int> values) async {
```

```
    final file = File(fileName);
```

```
    String content = "\$date Today \$end\n\n$timescale 1us \$end\n\n";
```

```
    for (int i = 0; i < timestamps.length; i++) {
```

```
        content += "$#{timestamps[i]}\n${values[i]}!\n";
```

```
    await file.writeAsString(content);}
```

4-Potential Challenges

•**Memory Usage:** I will start with a simple implementation and then improve it to handle larger files efficiently.

•**Time Synchronization:** Ensuring multiple channels remain aligned may be challenging, so I will include validation steps during testing.

•**Mobile Resource Management:** Handling large VCD files on mobile devices requires efficient memory management in Dart to prevent the app from lagging during data export

5-Community Engagement:

I have joined the PSLab Discord and introduced myself to the community. I have also started exploring the repository and looking through beginner-friendly issues to understand the project structure.

I plan to:

- Contribute small fixes or documentation improvements before the coding period.
- Stay active in discussions and ask questions when needed.
- Follow the project's coding standards and incorporate feedback from mentors.

I have been active on the PSLab Discord server, specifically in the #gsoc channel, to clarify project requirements so I initiated a discussion regarding the VCD implementation to ensure my approach aligns with the project's standards. You can find my query here:

<https://discord.com/channels/920089648842293248/942975979595395123/1486023498064199744>

6-Timeline:

- Weeks 1–2: Study PSLab Logic Analyzer data structures and understand the VCD format. And Deep dive into PSLab Android (Flutter) codebase and setting up the development environment for Dart.
- Weeks 3–6: Implement export functionality and test output using GTKWave.
- Weeks 7–9: Build the VCD parser for import functionality.
- Weeks 10–12: Testing, validation (export to import), and documentation.

7-Why This Project Matters to Me?

This project is an important step for me as I move into the tech field. It allows me to apply what I'm learning in Python while working with real data. I'm excited to learn from PSLab mentors and contribute something useful that helps users work more easily with their data.

Thanks for your time